MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

DATA ANALYTICS

MASTER PROGRAM

Turn Data into Smarter Decisions



COURSE HIGHLIGHTS

- Excel & Data Analysis
- Python for Data Analytics
- SQL & MySQL
- Power BI & Tableau
- Advanced Analytics & Capstone Projects

DATA SCIENCE WITH GEN AI

Learn • Analyze • Build • Innovate

COURSE OVERVIEW

A practical, career-focused program covering Excel, Python, SQL, MySQL, Power BI, Tableau, data visualization, advanced analytics, and real-world business intelligence projects.



WHAT WILL YOU LEARN?

EXCEL & DATA ANALYSIS

- Excel fundamentals & formulas
- Advanced Excel functions
- Pivot Tables & Pivot Charts
- Power Query & Power Pivot
- Interactive dashboards

PYTHON FOR ANALYTICS

- Python programming fundamentals
- NumPy & Pandas
- Data cleaning & manipulation
- Exploratory Data Analysis
- Data visualization

SQL & MYSQL

- SQL fundamentals
- Filtering & aggregation
- Joins & subqueries
- Advanced SQL
- Database design & optimization

BUSINESS INTELLIGENCE

- Power BI
- Tableau
- Data modeling
- DAX & calculated measures
- Interactive dashboards

ADVANCED ANALYTICS

- Machine Learning
- NLP
- Time Series
- APIs & data extraction
- Automation & data scraping

WHO CAN JOIN?

- Students & graduates
- Beginners in data analytics
- Aspiring data analysts
- Career changers
- Working professionals
- Business & management professionals

LEARNING OUTCOMES

By the end of the course, learners can:

- Analyze and clean real-world datasets
- Perform advanced Excel analysis
- Use Python for data analytics
- Query and manage MySQL databases
- Create interactive Power BI dashboards
- Build Tableau visualizations
- Perform EDA and statistical analysis
- Apply basic machine learning techniques
- Automate data analytics workflows
- Solve business problems using data

BUILD YOUR DATA SKILLS FOR A SMARTER TOMORROW

EXCEL & PYTHON FOR DATA ANALYTICS

Build Skills. Analyze Data. Make Better Decisions.

EXCEL FOR DATA ANALYSIS

Data Analysis Fundamentals

- Excel Basics
- Workbook & worksheet structure
- Cell references
- Relative, absolute & mixed references
- Data formatting
- Sorting & filtering
- Data validation

Formulas & Functions

- SUM • AVERAGE • COUNT
- MIN • MAX
- Logical & lookup functions
- VLOOKUP • HLOOKUP • XLOOKUP
- INDEX & MATCH

Data Visualization

- Bar & line charts
- Pie charts
- Sparklines
- Conditional formatting
- Interactive dashboards

ADVANCED EXCEL

- Business Data Analysis
- Pivot Tables & Pivot Charts
- Slicers & calculated fields
- Goal Seek & Solver
- Scenario Manager
- Descriptive statistics

POWER QUERY & POWER PIVOT

- Import data from multiple sources
- Clean & transform datasets
- Data relationships
- DAX fundamentals
- Advanced dashboards



PRACTICAL LEARNING

REAL-WORLD PROJECTS

EXPERT MENTORSHIP

CAREER GROWTH

TURN IDEAS INTO IMPACT

PYTHON FOR DATA ANALYTICS

PYTHON FUNDAMENTALS

- Syntax & variables
- Data types & operators
- Conditions & loops
- Functions & modules
- Lists • Tuples • Sets • Dictionaries

DATA ANALYSIS WITH PYTHON

NumPy

- Arrays
- Numerical operations
- Data processing

Pandas

- DataFrames
- Data selection & filtering
- Missing data handling
- GroupBy & aggregation
- Merge & join operations

DATA VISUALIZATION

- Matplotlib
- Seaborn
- Line charts
- Bar charts
- Scatter plots

KEY SKILLS

Collect → Clean → Transform → Analyze → Visualize

SQL, MYSQL & DATA VISUALIZATION

SQL & MYSQL FOR DATA ANALYTICS

SQL FUNDAMENTALS

- SELECT
- WHERE
- ORDER BY
- LIMIT
- AND • OR • IN
- NULL handling
- Wildcard searches

DATA AGGREGATION

- COUNT
- SUM
- AVG
- MIN • MAX
- GROUP BY
- HAVING

SQL JOINS

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- Self Join
- CROSS JOIN

ADVANCED SQL

- Subqueries
- Window functions
- RANK()
- ROW_NUMBER()
- LEAD() • LAG()
- Stored procedures
- Triggers

DATABASE MANAGEMENT

- Primary & foreign keys
- Relationships
- Normalization
- Indexing
- Transactions
- COMMIT & ROLLBACK
- Query optimization
- EXPLAIN plans

KEY HIGHLIGHTS

Write • Analyze • Manage • Optimize

POWER BI FOR DATA VISUALIZATION

DATA CONNECTION

- Excel
- SQL
- Web data
- Multiple data sources

POWER QUERY

- Data cleaning
- Transformation
- Filtering
- Data preparation

DATA MODELING

- Relationships
- Calculated columns
- Measures
- DAX fundamentals

DASHBOARDS & REPORTS

- Charts & tables
- Slicers
- KPI indicators
- Dynamic filters
- Custom visuals
- Interactive dashboards

PUBLISH & SHARE

- Report publishing
- Dashboard sharing
- Stakeholder reporting

KEY HIGHLIGHTS

Connect • Transform • Analyze • Share

BUILD MODERN SKILLS

PRACTICAL HANDS-ON LEARNING

INDUSTRY RELEVANT SKILLS

GROW YOUR CAREER

SKILLS TODAY
FOR A BRIGHTER TOMORROW

TABLEAU & ADVANCED DATA ANALYTICS

TABLEAU FOR DATA VISUALIZATION

DATA CONNECTION
- Excel
- SQL
- Multiple data sources
- Data relationships

VISUAL ANALYTICS
- Bar charts
- Pie charts
- Line charts
- Histograms
- Maps

ADVANCED TABLEAU
- Calculated fields
- Table calculations
- Data blending
- Joins & relationships
- Filters & parameters
- Dashboard actions

INTERACTIVE DASHBOARDS
Connect → Analyze → Visualize → Interact → Share

- Interactive dashboards
- Dynamic filters
- Parameters
- Dashboard navigation
- Publishing & sharing

PYTHON FOR ADVANCED DATA ANALYTICS

ADVANCED DATA HANDLING
- Hierarchical & multi-level data
- APIs & data extraction
- Advanced data manipulation
- Automated data processing

MACHINE LEARNING
- Scikit-learn
- Regression
- Classification
- Clustering
- Model evaluation
- Cross-validation
- Hyperparameter tuning

NATURAL LANGUAGE PROCESSING
- Text preprocessing
- Tokenization
- Text cleaning
- Sentiment analysis

TIME SERIES ANALYTICS
- Time series fundamentals
- ARIMA
- Prophet
- Forecasting models

ADVANCED ANALYTICS

Explore → Model → Evaluate → Predict → Communicate

BUILD MODERN WEB APPLICATIONS

DEVELOP SCALABLE SOLUTIONS

GAIN PRACTICAL SKILLS

BECOME INDUSTRY-READY

SKILLS TODAY
FOR A BRIGHTER TOMORROW

DATA SCIENCE WITH GEN AI

REAL-WORLD PROJECT, INTEGRATION & CAREER SKILLS

DATA INTEGRATION & REAL-WORLD APPLICATION

Excel → Python → MySQL → Power BI / Tableau → Business Insights

ADVANCED VISUALIZATION

- Plotly
- Bokeh
- Matplotlib
- Interactive analytics

AUTOMATION & DATA EXTRACTION

- Python automation
- Data scraping
- API-based data collection
- Automated reports
- Scheduled analytics workflows

CLOUD & BIG DATA OVERVIEW

- AWS
- Microsoft Azure
- Google Cloud
- Hadoop
- Apache Spark

CAPSTONE PROJECT

SOLVE A REAL BUSINESS PROBLEM WITH DATA

PROJECT PROCESS

Define → Collect → Clean → Analyze → Visualize → Present

PROJECT COMPONENTS

DATA COLLECTION

- Excel datasets
- SQL / MySQL
- Python
- APIs & external data

DATA ANALYSIS

- Data cleaning
- EDA
- Statistical analysis
- Business insights

VISUALIZATION

- Power BI dashboards
- Tableau dashboards
- Interactive reports

ADVANCED ANALYTICS

- Machine learning
- Forecasting
- Predictive analysis

BUSINESS PRESENTATION

- Key findings
- Data-driven insights
- Recommendations
- Stakeholder presentation

TOOLS & TECHNOLOGY STACK

Excel | Power Query | Power Pivot | Python | NumPy | Pandas | MySQL | SQL | Power BI | Tableau | Scikit-learn | Matplotlib | Seaborn

FINAL OUTCOME

Collect → Clean → Analyze → Visualize → Predict → Communicate

YOUR DATA SCIENCE JOURNEY STARTS HERE.



PRACTICAL LEARNING

REAL-TIME SCENARIOS

HANDS-ON PROJECTS

CAREER GROWTH

SKILLS TODAY FOR A SMARTER TOMORROW